Assignment 3: Data Exploration

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Spring 2026

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. [NEW] Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to Canvas.
8. Initial here to acknowledge that you did not use AI in completing this assignment, except where expressly allowed: _Yiting He___

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks in your code chunks.

TIP: If your code fails to knit, check: * That no `install.packages()` or `View()` commands exist in your code. * That you are not displaying the entire contents of a large dataframe in your code.

Set up your R session

1. Load necessary packages (tidyverse, here), check your current working directory and import two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets "Neonics" and "Litter", respectively.

Be sure to: * Use the `here()` package in specifying the paths to your datasets * Include the appropriate subcommand to read in character based columns as factors

```r
# Load required packages
library(tidyverse)
library(here)

# Check working directory
getwd()
```

```
## [1] "/home/guest/R/ENVIOR-872/EDE_Spring2026/Data/Metadata"
```

```r
# Import datasets using corrected paths (Data > Raw)
Neonics <- read.csv(
  here("Data", "Raw", "ECOTOX_Neonicotinoids_Insects_raw.csv"),
  stringsAsFactors = TRUE
)

Litter <- read.csv(
  here("Data", "Raw", "NEON_NIWO_Litter_massdata_2018-08_raw.csv"),
  stringsAsFactors = TRUE
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information. (AI is allowed here, but put answers in your own words.)

   Answer: Neonicotinoids are systemic insecticides that are widely used but controversial due to their non-target effects. We are interested in their ecotoxicology because they are linked to the decline of essential pollinators like honeybees and bumblebees. Understanding their impact helps in assessing risks to biodiversity, food security, and the health of beneficial insect populations.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information. (AI is allowed here, but put answers in your own words.)

   Answer:Litter and woody debris are critical components of nutrient cycling and carbon storage in forest ecosystems. Studying them helps scientists understand the rate of primary production, decomposition processes, and how energy is transferred from the forest canopy to the soil. It also provides insights into how forest health responds to climate change.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

   Answer: 1.Litter is collected using both elevated leaf traps and ground-placed traps (for larger woody debris). 2.Sampling occurs in designated "tower plots" to allow for integration with atmospheric and soil measurements. 3. Collected samples are sorted by functional groups (e.g., leaves, needles, twigs) and dried to obtain a constant dry mass.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```r
# Check dimensions of the neonicotinoid dataset
dim(Neonics)
```

```
## [1] 4623   30
```

6. Using the `summary` function on the "Effect" column, determine the most common effects that are studied. [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```r
# Summarize and sort Effect column
effect_counts <- summary(Neonics$Effect)
sort(effect_counts, decreasing = TRUE)
```

```
##       Population        Mortality         Behavior Feeding behavior 
##             1803             1493              360              255 
##     Reproduction      Development        Avoidance         Genetics 
##              197              136              102               82 
##        Enzyme(s)           Growth       Morphology    Immunological 
##               62               38               22               16 
##     Accumulation     Intoxication     Biochemistry          Cell(s) 
##               12               12               11                9 
##       Physiology        Histology       Hormone(s) 
##                7                5                1
```

Question: Which two effects stand out as the most studied? Can you guess why these effects might specifically be of interest? > Answer:Mortality and growth/reproduction-related effects typically stand out as the most studied. Mortality is a direct and easily measurable endpoint, while growth and reproduction are critical for population persistence, making them especially relevant for ecological risk assessment.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name).[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```r
# Identify most commonly studied species using summary with maxsum
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##            Honey Bee       Parasitic Wasp Buff Tailed Bumblebee 
##                  667                  285                   183 
##   Carniolan Honey Bee           Bumble Bee      Italian Honeybee 
##                  152                  140                   113 
##              (Other) 
##                 3083
```

Question: What do these species have in common? Why might they be of interest over other insects? > Answer:These species are often model organisms or economically/ecologically important insects, such as pollinators or pest species. They are well-studied, relatively easy to culture in laboratory settings, and their responses are relevant for both agricultural productivity and environmental protection.

8. The Conc.1..Author column, which lists the concentration of the neonicitoid dose, should include numeric values. What is the class of Conc.1..Author. column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
# Check class of concentration column
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

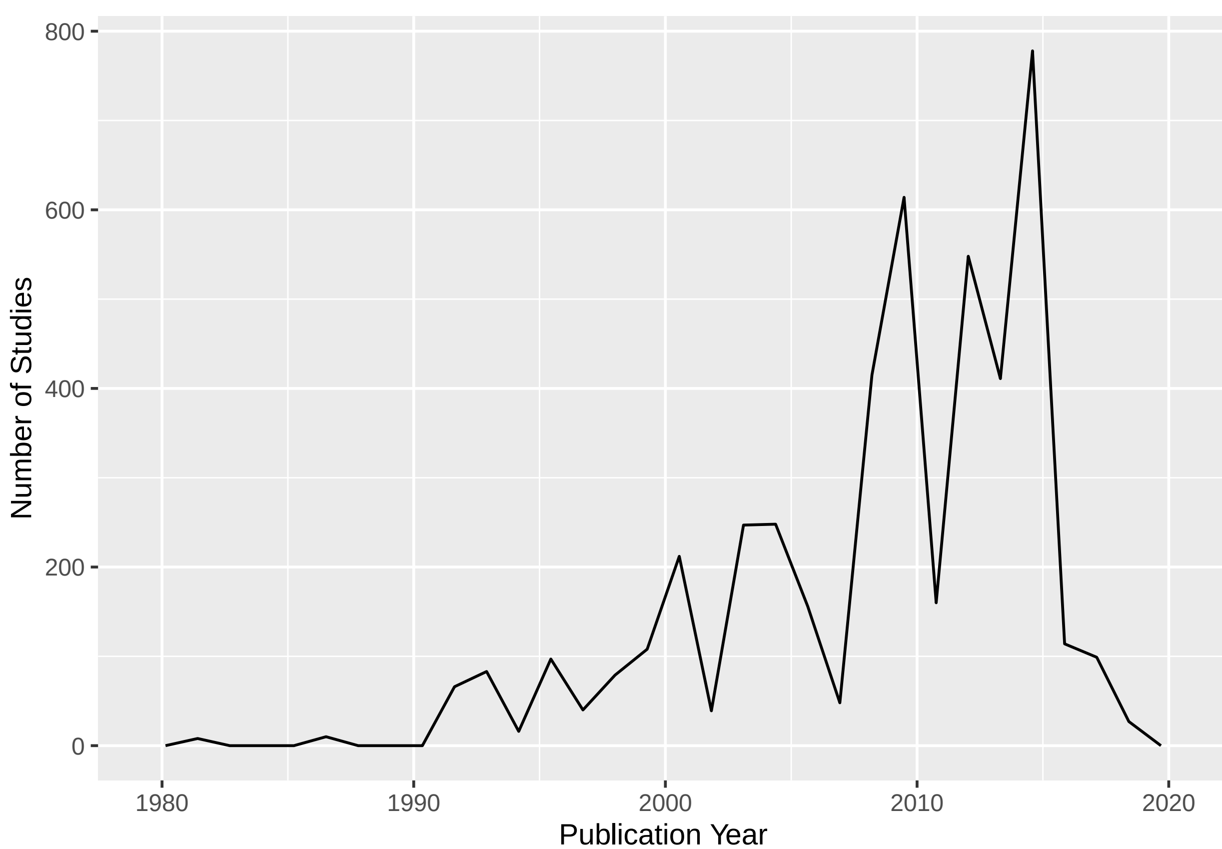
Answer:The column is typically stored as a factor or character rather than numeric because it may contain non-numeric entries such as ranges, text qualifiers, or units. These mixed formats prevent R from interpreting the column as purely numeric.

Explore your data graphically (Neonics)

9. Using geom_freqpoly, generate a plot of the number of studies conducted by publication year.

```
# Plot number of studies by publication year
ggplot(Neonics, aes(x = Publication.Year)) +
geom_freqpoly() +
labs(x = "Publication Year", y = "Number of Studies")
```

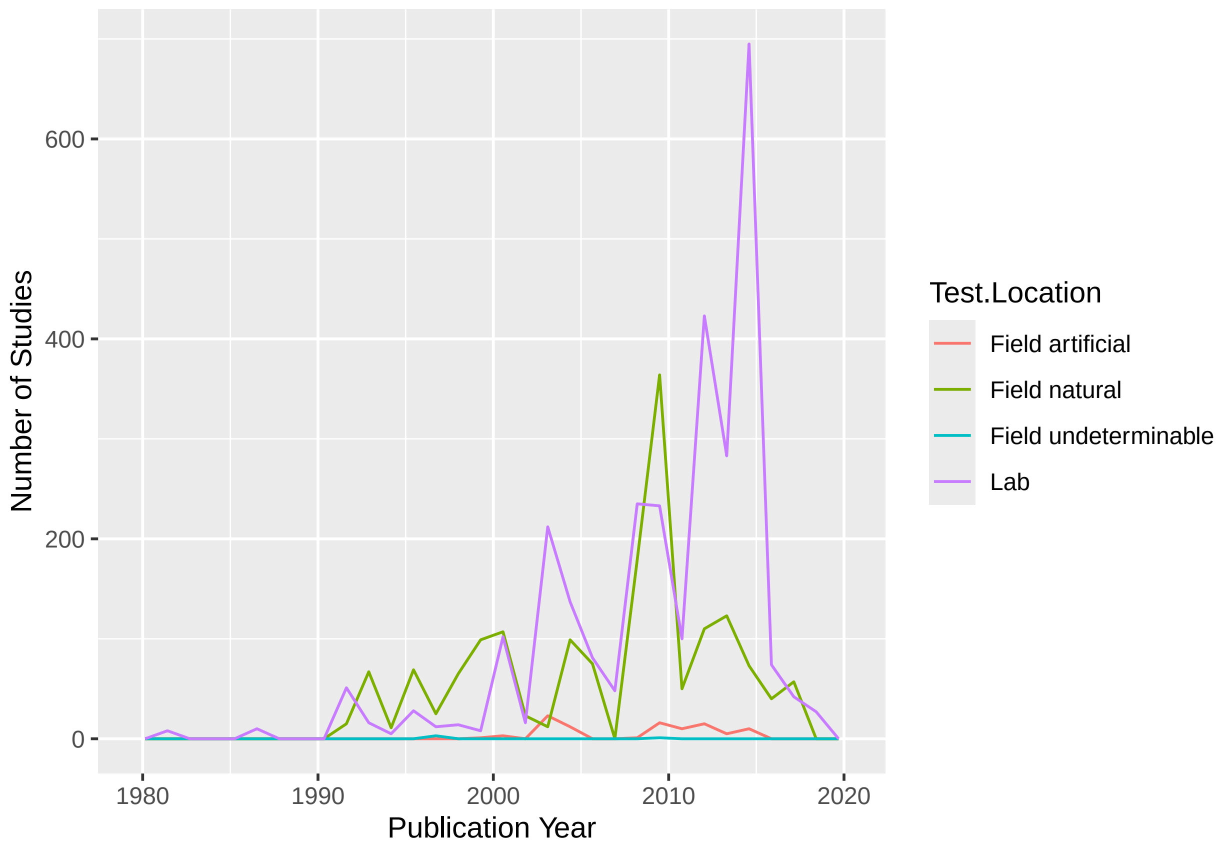
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
# Plot frequency polygon by test location
ggplot(Neonics, aes(x = Publication.Year, color = Test.Location)) +
geom_freqpoly() +
labs(x = "Publication Year", y = "Number of Studies")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

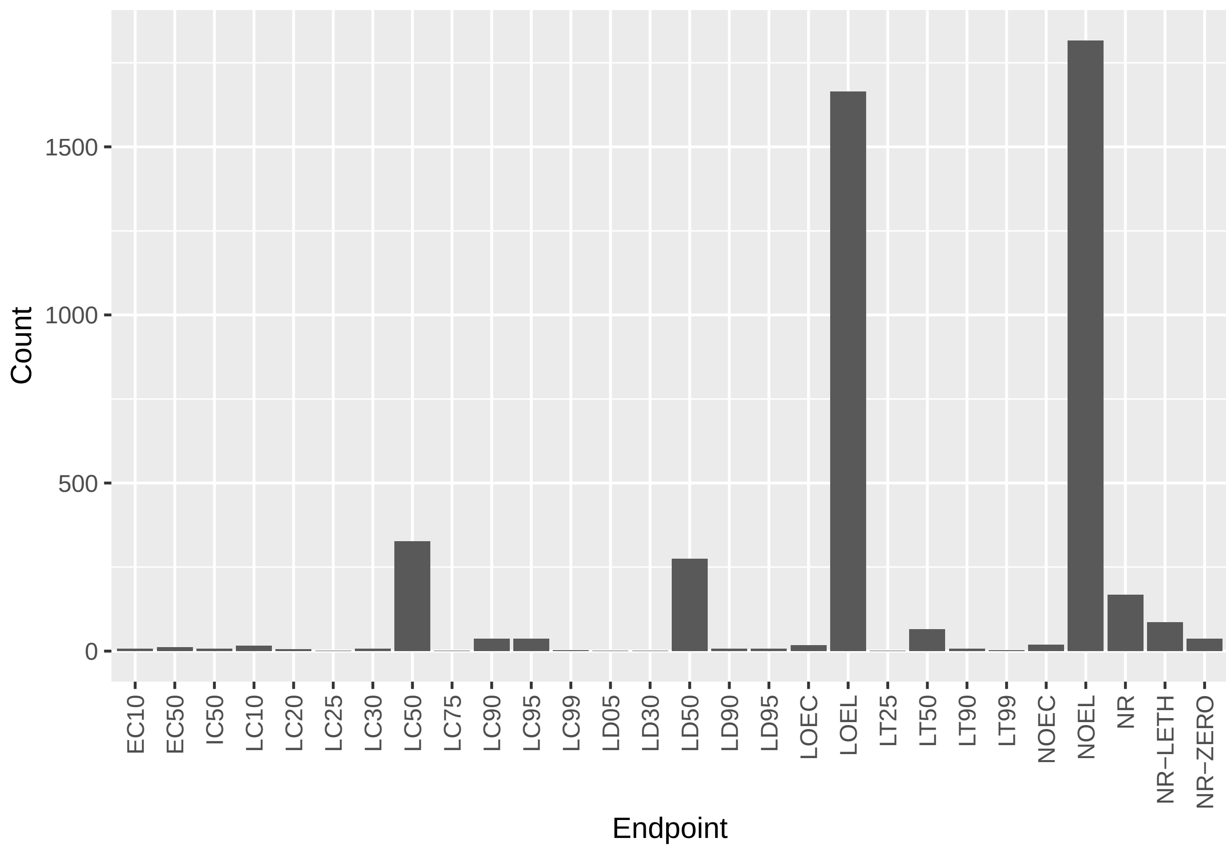


Interpret this graph. What are the most common test locations, and do they differ over time? > Answer:Laboratory studies are generally the most common test location, reflecting greater experimental control and reproducibility. Field and semi-field studies appear less frequently but may increase over time as interest grows in ecological realism and real-world exposure scenarios.

11. Create a bar graph of Endpoint counts.

[**TIP**: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels…]

```
# Bar graph of endpoint counts
ggplot(Neonics, aes(x = Endpoint)) +
geom_bar() +
theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)) +
labs(x = "Endpoint", y = "Count")
```



What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix (p.721) for more information. > Answer:Mortality and growth/reproduction endpoints are most common. Mortality refers to death of the organism following exposure, while growth or reproduction endpoints measure sub-lethal effects such as reduced development, fecundity, or emergence success.

Explore your data (Litter)

12. Determine the class of `collectDate`. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
# Check class of collectDate
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
# Convert to Date
Litter$collectDate <- as.Date(Litter$collectDate)


# Confirm new class
class(Litter$collectDate)
```

```
## [1] "Date"
```

```r
# Unique sampling dates in August 2018
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different `plotIDs` sampled at Niwot Ridge.

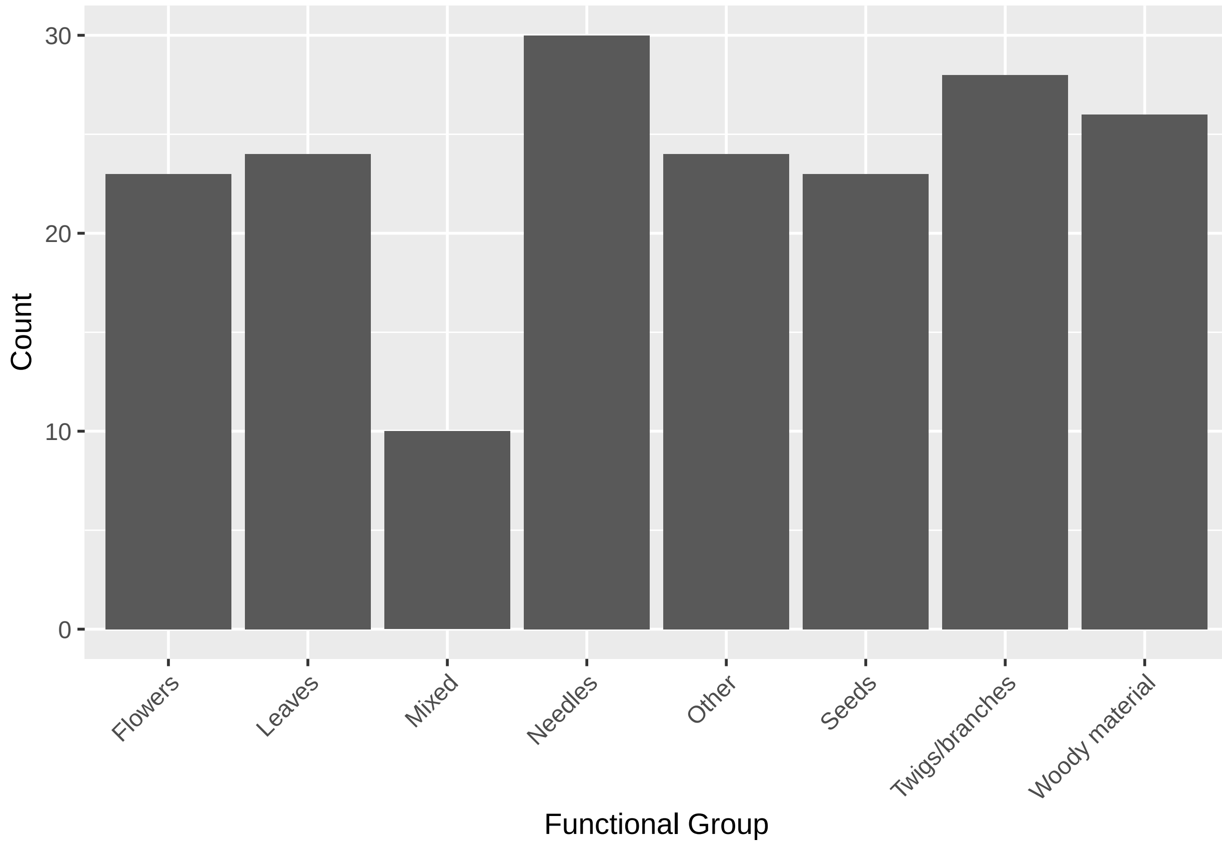
```r
# List unique plot IDs
unique(Litter$plotID)
```

```
##  [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
##  [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

How is the information obtained from `unique` different from that obtained from `summary`? > Answer:The unique function lists each distinct value only once, whereas summary provides counts or descriptive statistics, showing how frequently each value occurs.

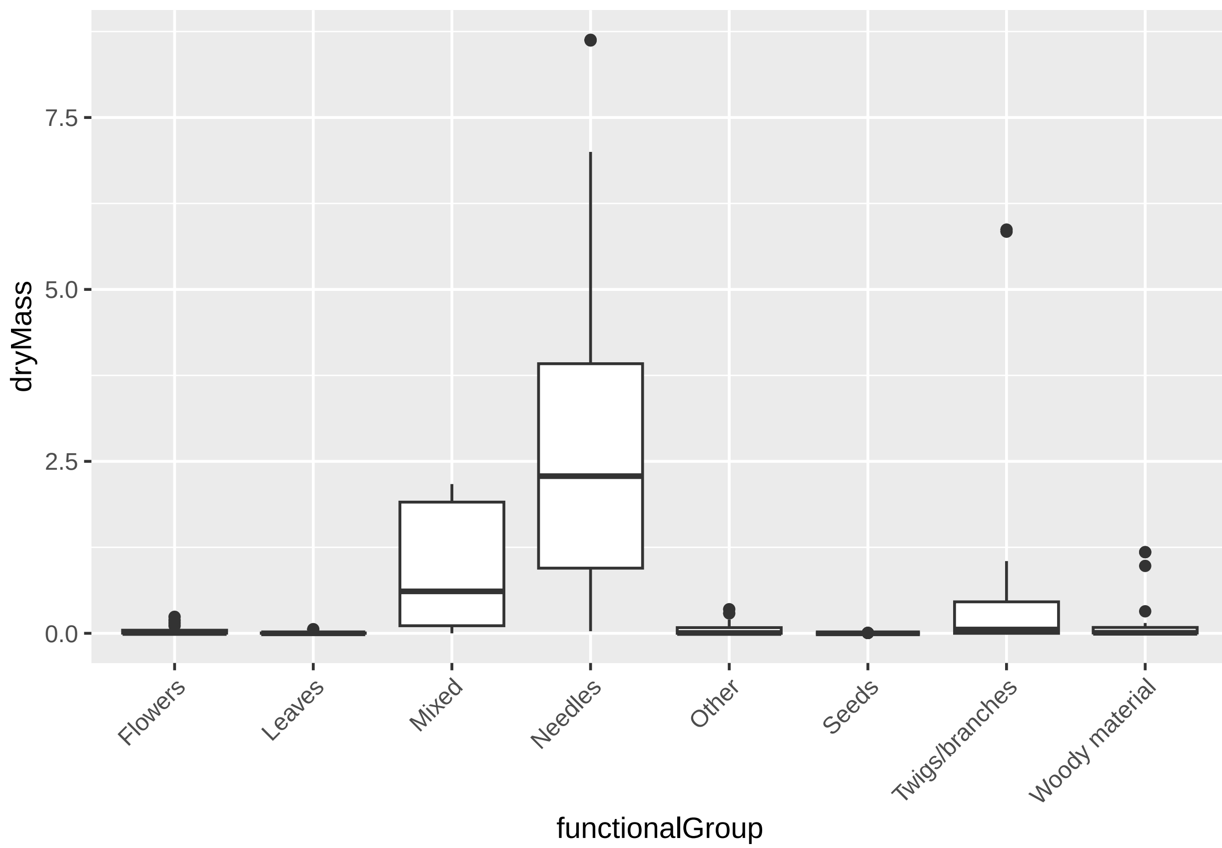
14. Create a bar graph of `functionalGroup` counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```r
# Bar graph of functional group counts
ggplot(Litter, aes(x = functionalGroup)) +
geom_bar() +
theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
labs(x = "Functional Group", y = "Count")
```

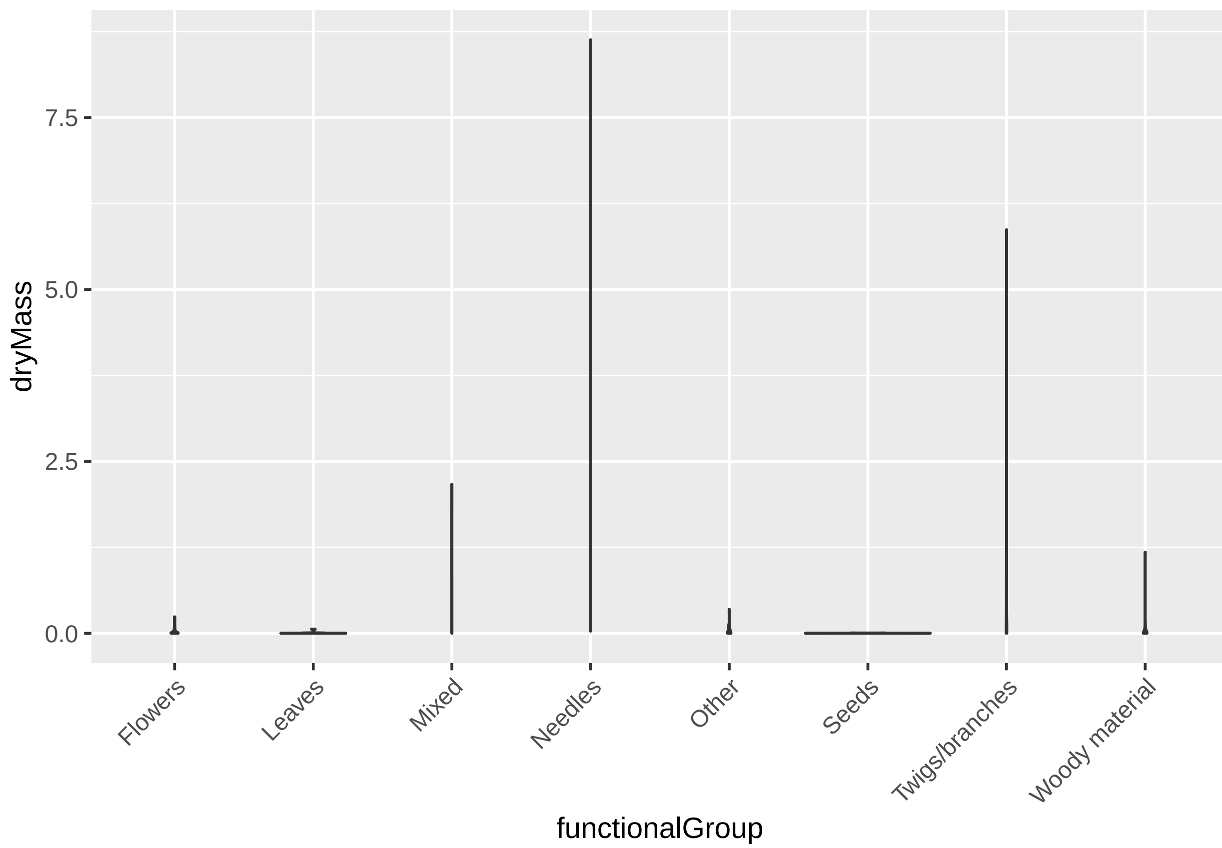


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
# Boxplot
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
geom_boxplot() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
# Violin plot
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
geom_violin() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Why is the boxplot a more effective visualization option than the violin plot in this case? > Answer:The boxplot more clearly summarizes the median, interquartile range, and outliers. With limited data points per group, the violin plot can give a misleading impression of distribution shape.

What type(s) of litter tend to have the highest biomass at these sites? > Answer:Woody debris and larger leaf or needle litter types tend to have the highest biomass, reflecting their greater mass per unit compared to finer litter components.